

How to Map Any Formal System to $V / G / \theta$

A six-step identification guide for adding carriers to the library

Every formal system ever built is a specific choice of three parameters: a value space V , a gradient family G , and a coherence threshold θ . The propagation operator $P/G \rightarrow Q$ is constant across all of them. Follow these six steps to identify V , G , and θ for any system, derive its forced laws by enumeration, find its failures by counterexample, and locate its morphisms to other carriers.

-
- 1 Identify V — the value space
 - 2 Identify G — the gradient family
 - 3 Identify θ — the coherence threshold
 - 4 Derive forced laws by enumeration
 - 5 Find failures by counterexample
 - 6 Identify morphisms to other carriers
-

$P / G \rightarrow Q$ — the operator is constant. Only V , G , θ vary.

The mechanism is fixed. The carrier is chosen. The rules follow.

1 Identify V — the value space

What are the possible outputs or states the system produces? If the system assigns truth values, V might be $\{0,1\}$ or $[0,1]$ or a finite set of labels. If it assigns probabilities, $V=[0,1]$ (probability distributions). If it assigns positions in a space, V is that space. Ask: “What does the system return or produce?”

EXAMPLES

Classical logic: $V=\{0,1\}$. Fuzzy logic: $V=[0,1]$. Quantum states: $V=\text{Hilbert space } H$. Ordinals: $V=\text{Ord}$.

2 Identify G — the gradient family

What operations or transformations are available? These are the generators that the propagation operator uses to move from premises to conclusions. They may be logical connectives, mathematical operations, field operators, or algebraic morphisms. Ask: “What can be done to elements of V ?”

EXAMPLES

Classical: $G=\{\text{NOT, AND, OR}\}$. Probability: $G=\{\text{conditional expectation, Bayes update}\}$. Lie groups: $G=\{\text{Lie bracket, exp map}\}$. Category theory: $G=\{\text{functor application, natural transformation composition}\}$.

3 Identify θ — the coherence threshold

What counts as ‘full truth’, ‘validity’, or ‘acceptance’ in this system? This is the boundary value that separates coherent results from incoherent ones. It may be a fixed value (1, or ω , or 0), a condition (non-degeneracy of the Killing form), or a structural property (sigma-additivity). Ask: “At what value or condition does the system accept a result as valid?”

EXAMPLES

Classical: $\theta=1$ (truth). Probability: $\theta=P(Q)=1$. Fuzzy Gödel: $\theta=1$ (only crisp 1 is fully true). Symplectic: $\theta=\text{Liouville conservation}$. Ordinals: $\theta=\epsilon_0$ (consistency strength of PA).

4 Derive forced laws by enumeration

For finite V : exhaustively check every combination of inputs under every G operator. A law is forced if it holds for ALL combinations with no exception. This is mechanical verification that can be automated. For infinite/continuous V : verify algebraic identities symbolically (e.g., $d^2=0$ for exterior derivative, $[X,Y]=-[Y,X]$ for Lie bracket). Ask: “Which properties follow purely from the arithmetic of V and G ?”

EXAMPLES

Classical: $\text{NOT}(\text{NOT}(v))=v$ for $v \in \{0,1\}$ checked by two cases. LNC: $v \text{ AND } \text{NOT}(v)=0$ for both $v=0$ and $v=1$. Itô: quadratic variation $[W,W]_t=t$ forces the correction term in Itô’s lemma.

5 Find failures by counterexample

For each law expected from analogy with another carrier, seek a specific input combination that violates it. A single counterexample is sufficient to establish that the law does not hold. The counterexample reveals the exact arithmetic reason for the failure. Ask: “Which classical laws does this carrier violate, and where exactly does the computation fail?”

EXAMPLES

L3 logic: LNC fails at $v=1/2$ because $\text{AND}(1/2, \text{NOT}(1/2)) = \min(1/2, 1/2) = 1/2 \neq 0$. Many-worlds: collapse fails because unitarity forces all branches to persist. Ordinals: commutativity fails because $1+\omega=\omega$ but $\omega+1>\omega$.

6 Identify morphisms to other carriers

A morphism is a map $f: V \rightarrow V'$ that commutes with all G operations: $f(G(v)) = G'(f(v))$. Morphisms reveal structural relationships between carriers. A bijective morphism is an isomorphism (the two carriers are ‘the same’ PL mechanism in different notation). Ask: “Is there a map to another carrier that preserves all G operations?”

EXAMPLES

Classical $\rightarrow$ Probability: inclusion $\{0,1\} \subset [0,1]$ is a morphism (every classical value is a valid probability). Gödel $\rightarrow$ classical at $\theta=1$: the thresholding map $v \mapsto [v=1]$ is a morphism. Every probability measure is a morphism from Boolean events to $[0,1]$.

VERIFICATION PRINCIPLE

Every forced law must be re-derivable from V , G , and θ alone by the procedure above. If a law cannot be derived this way, it is either an additional axiom being imposed (and the carrier should note it as such) or the $V/G/\theta$ specification is incomplete. Run `library_verify.py` on the Code page to see automated verification for all carriers in the library.